INJECTIVE QUASI-COHERENT MODULES ON AN AFFINE NOETHERIAN SCHEME ARE FILTERED COLIMITS OF ARTINIAN PUSHFORWARDS

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Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file spaces-perfect.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: the proof reduces to indecomposable injectives via the structure theorem, writes the injective hull as the union of its Artinian layers using the union lemma, and reads the layers as pushforwards of coherent sheaves from Artinian spectra

Editorial note. The label lemma-affine-injective-colimit-direct-sum-pushforwards-artin names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own lemma environment in the chapter source.

Allowed standard prerequisites (editorial).

- the equivalence between QCoh(O_S) on an affine scheme and the category of A-modules
- the structure theorem for injective modules over a Noetherian ring: every injective is a direct sum of indecomposable injectives, each the injective hull of a residue field (Dualizing Complexes, Proposition dualizing-proposition-structure-injectives-noetherian)
- the description of the injective hull of kappa(p) as the union of its Artinian submodules J[p^n], the injective hulls over A_p/p^n A_p (Dualizing Complexes, Lemma dualizing-lemma-union-artinian)
- finiteness of the pieces (Dualizing Complexes, Lemma dualizing-lemma-finite)

Source reference. Original lemma 08YX: sources/stacks-chow/tags.txt, source lines 11355–11355. Prerequisite pointer invoked at proof line(s) 2864; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 08Z4: sources/stacks-chow/tags.txt, source lines 11362–11362. Prerequisite pointer invoked at proof line(s) 2857; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original proposition 08YA: sources/stacks-chow/tags.txt, source lines 11333–11333. Prerequisite pointer invoked at proof line(s) 2848; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Date: stacks-project:spaces-perfect.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Lemma 0.1. *Let S be a Noetherian affine scheme. Every injective object of $QCoh(\mathcal{O}_S)$ is a filtered colimit $\operatorname{colim}_i \mathcal{F}_i$ of quasi-coherent sheaves of the form*

$$\mathcal{F}_i = (Z_i \to S)_* \mathcal{G}_i$$

where Z_i is the spectrum of an Artinian ring and $\mathcal{G}_i$ is a coherent module on Z_i.

Proof. Let $S = \operatorname{Spec}(A)$. Let $\mathcal{J}$ be an injective object of $QCoh(\mathcal{O}_S)$. Since $QCoh(\mathcal{O}_S)$ is equivalent to the category of A-modules we see that $\mathcal{J}$ is equal to $\widetilde{J}$ for some injective A-module J. By Dualizing Complexes, Proposition 08YA we can write $J = \bigoplus E_\alpha$ with E_α indecomposable and therefore isomorphic to the injective hull of a reside field at a point. Thus (because finite disjoint unions of Artinian schemes are Artinian) we may assume that J is the injective hull of $\kappa(\mathfrak{p})$ for some prime $\mathfrak{p}$ of A. Then $J = \bigcup J[\mathfrak{p}^n]$ where $J[\mathfrak{p}^n]$ is the injective hull of $\kappa(\mathfrak{p})$ over $A_{/}\mathfrak{p}^n A_\mathfrak{p}$, see Dualizing Complexes, Lemma 08Z4. Thus $\widetilde{J}$ is the colimit of the sheaves $(Z_n \to X)_* \mathcal{G}_n$ where $Z_n = \operatorname{Spec}(A_\mathfrak{p}/\mathfrak{p}^n A_\mathfrak{p})$ and $\mathfrak{G}_n$ the coherent sheaf associated to the finite $A_{/}\mathfrak{p}^n A_\mathfrak{p}$-module $J[\mathfrak{p}^n]$. Finiteness follows from Dualizing Complexes, Lemma 08YX. □